Piush Kole

Data Science Student · Applying for CERN Short-Term Internship — Data Science, AI & Analytics

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SUMMARY

MSc Data Science candidate (CHRIST University, Bengaluru), currently a full-time student, applying for CERN's Short-Term Internship Programme under Data Science, AI & Analytics. Built and publicly documented an independent machine-learning research project — a reproducible benchmark of neural-network quantization — covering the full practical skill set a data-science placement draws on: Python/PyTorch model development, statistical evaluation, automated testing, and clear written documentation of methods and limitations. Comfortable picking up new tools and methods quickly (this project required learning `hls4ml`, a HEP-specific ML-deployment toolchain, from scratch), and used to working precisely and communicating findings clearly — including where a result went against the simplest narrative. Eager to apply this foundation to a real project under supervision, and to learn from CERN's own data-science and computing teams.

TECHNICAL SKILLS

- **ML & data science:** Python, PyTorch, scikit-learn, pandas, NumPy — model training, evaluation, and statistical analysis (multi-seed experimental design, confidence intervals)
- **Tools picked up for this project:** `hls4ml` (HEP-specific ML-deployment toolchain, learned from its own documentation), FastAPI, Docker, Git, pytest, CI (GitHub Actions)
- **Data & visualization:** SQL, Power BI, Plotly, Matplotlib
- **Working habits:** writing a selection rule before touching results, automated regression testing, documenting what a result does and doesn't support

PROJECT — HEP-TRIGGER QUANTIZATION LAB (INDEPENDENT)

github.com/Piush2004/hep-trigger-quantization-lab — public repository, code, and reports

An independent research project benchmarking how much a neural network's numerical precision can be reduced (a compression technique called quantization) before its accuracy on a classification task breaks down, using a public physics dataset.

- Built and evaluated multiple model variants (standard precision vs. three reduced-precision variants, two training strategies) across 5 random seeds each, with the comparison rule decided *before* looking at the held-out test results — to avoid picking whichever number looked best after the fact.
- Learned and applied `hls4ml`, a specialized toolchain used in HEP computing to convert a trained PyTorch model into fixed-point arithmetic for hardware deployment, and validated that its output numerically agreed with the original model.
- Found and reported a result that didn't match the simplest expectation: a standard, non-neural baseline model actually matched or beat the neural network on raw accuracy — reported plainly rather than left out.
- Wrote 141 automated tests (run twice independently to confirm identical results) and full reproduction documentation, so every number in the project's reports can be independently checked.

EDUCATION

M.Sc. Data Science (Computer Science) — CHRIST (Deemed to be University), Bengaluru *Jul 2025 – May 2027 (in progress, full-time)*

B.Sc. (Hons.) Data Science — Techno India University, Kolkata · *May 2022 – Apr 2025* · 69.67%

EXPERIENCE

Data Executive Intern — Cybertoss Pvt. Ltd., Kolkata (onsite) · *Jun 2024 – May 2025*

- Owned the end-to-end data lifecycle (collection, cleaning, validation, reporting) for large-scale datasets, automating Python workflows that reduced manual effort for cross-functional analytics.

Data Science Summer Intern — RemarkSkill (KSHITIJ, IIT Kharagpur), remote · *Jun 2024*

- Applied Python (pandas, NumPy, Matplotlib) and Power BI to real-world dataset projects.

CERTIFICATIONS & LEADERSHIP

AWS Cloud Practitioner Essentials (AWS Academy, 2025) · Python for Data Science & Machine Learning (Udemy, 2023) · IEEE Committee Member, CHRIST University (Dec 2025) — coordinated outreach to 10+ industry companies for a 20-member student-conference committee, developing communication and teamwork skills in a multicultural setting.